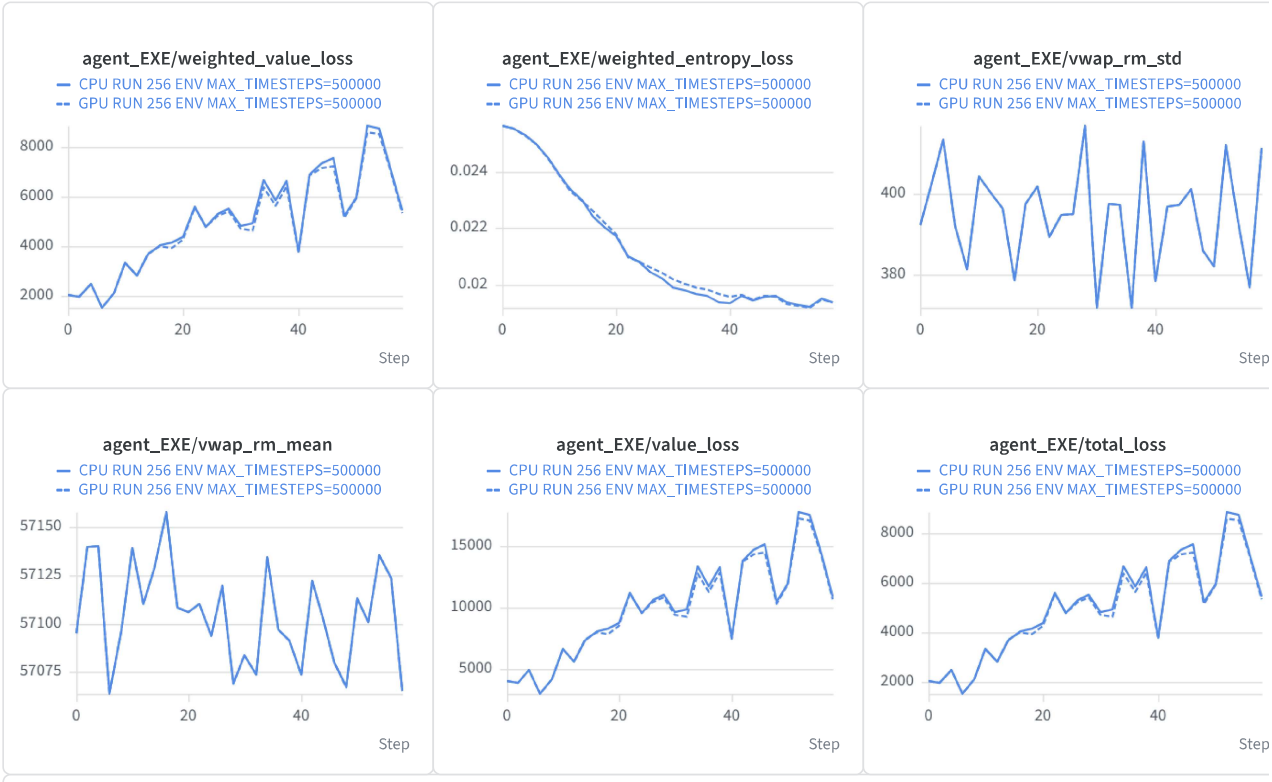
Comparison benchmark of 256 env GPU vs CPU

It was observed that CPU outperforms GPU at low environment counts (NUM_ENVS=256). However, as NUM_ENVS increases, CPU throughput saturates — limited by the number of physical cores — while GPU utilization increases proportionally with the number of parallel environments. This allows the GPU to exploit its thousands of CUDA cores, reducing total training time significantly at higher environment counts.

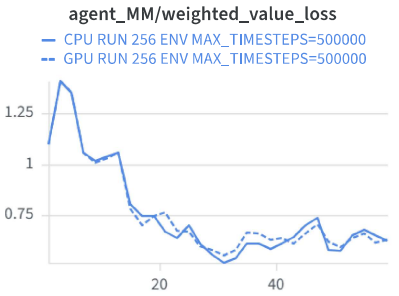
Duvvuri Venkata Naga Sai Ritvik me25b039

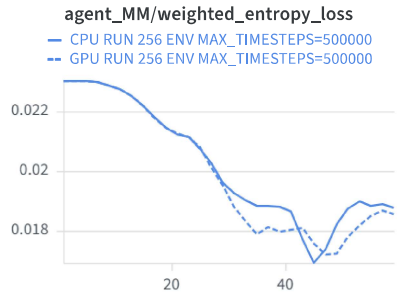
NOTE : All these Simulations were ran on a Laptop with intel i7-13700HX processor and Nvidia RTX 4060 GPU

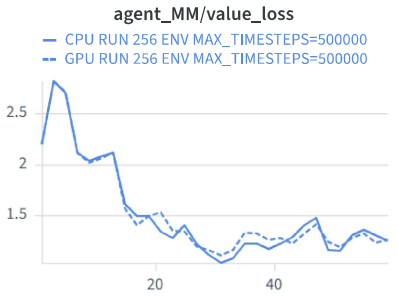
EXE Agent Metrics

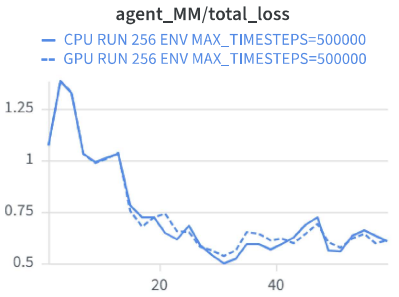


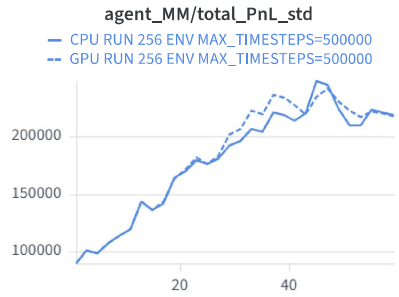
There are minor deviations Probably because of floating point Errors

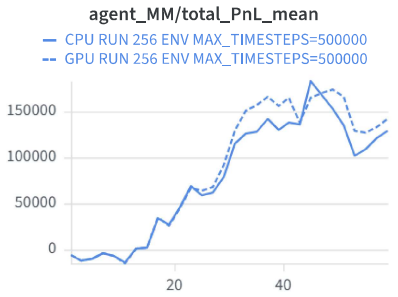




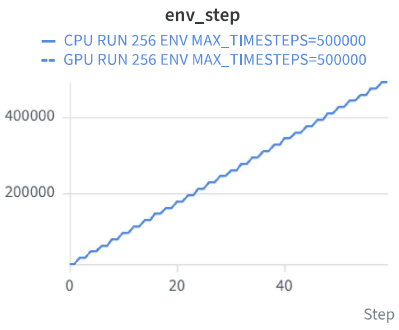








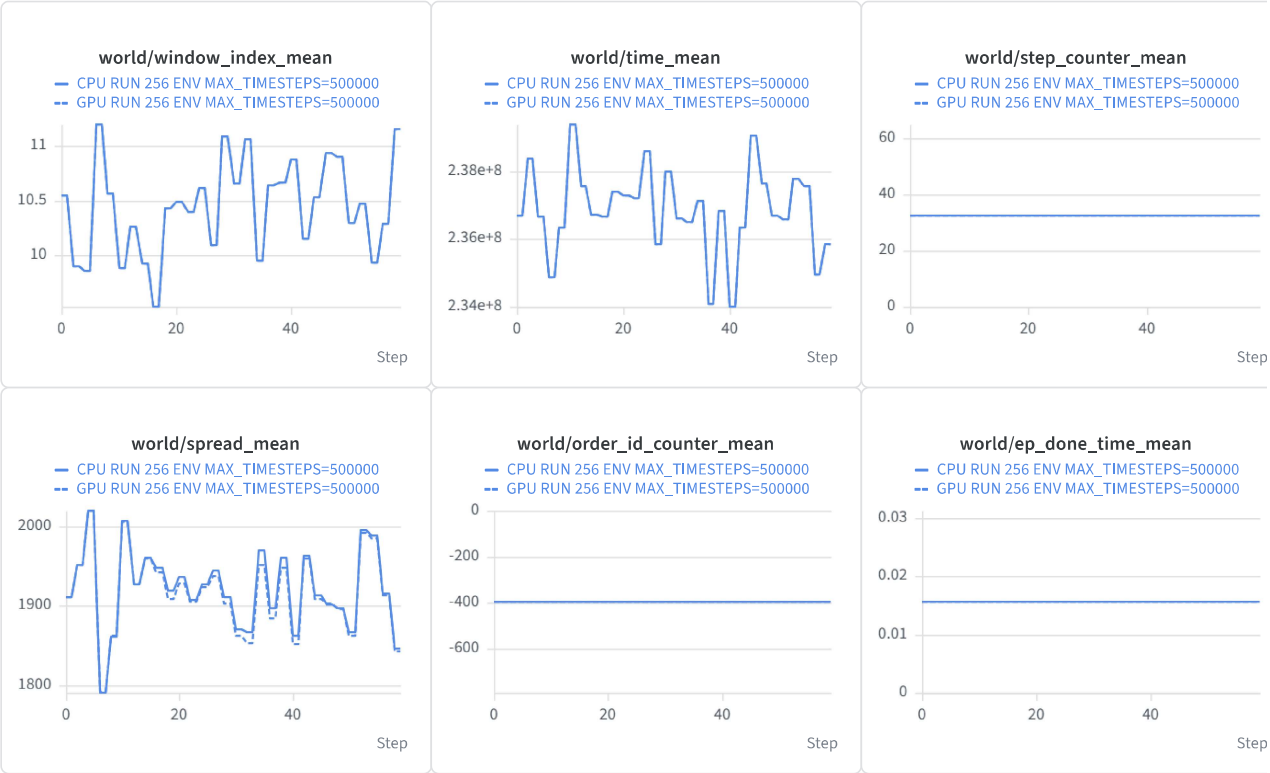
The MM agent demonstrates successful policy learning across both hardware configurations, with total_PnL_mean rising from near zero to approximately 150,000 by training completion. Loss metrics decrease consistently across 50 update steps, confirming convergence. CPU and GPU runs produce statistically identical training trajectories, validating that hardware choice affects only computational efficiency, not training outcomes



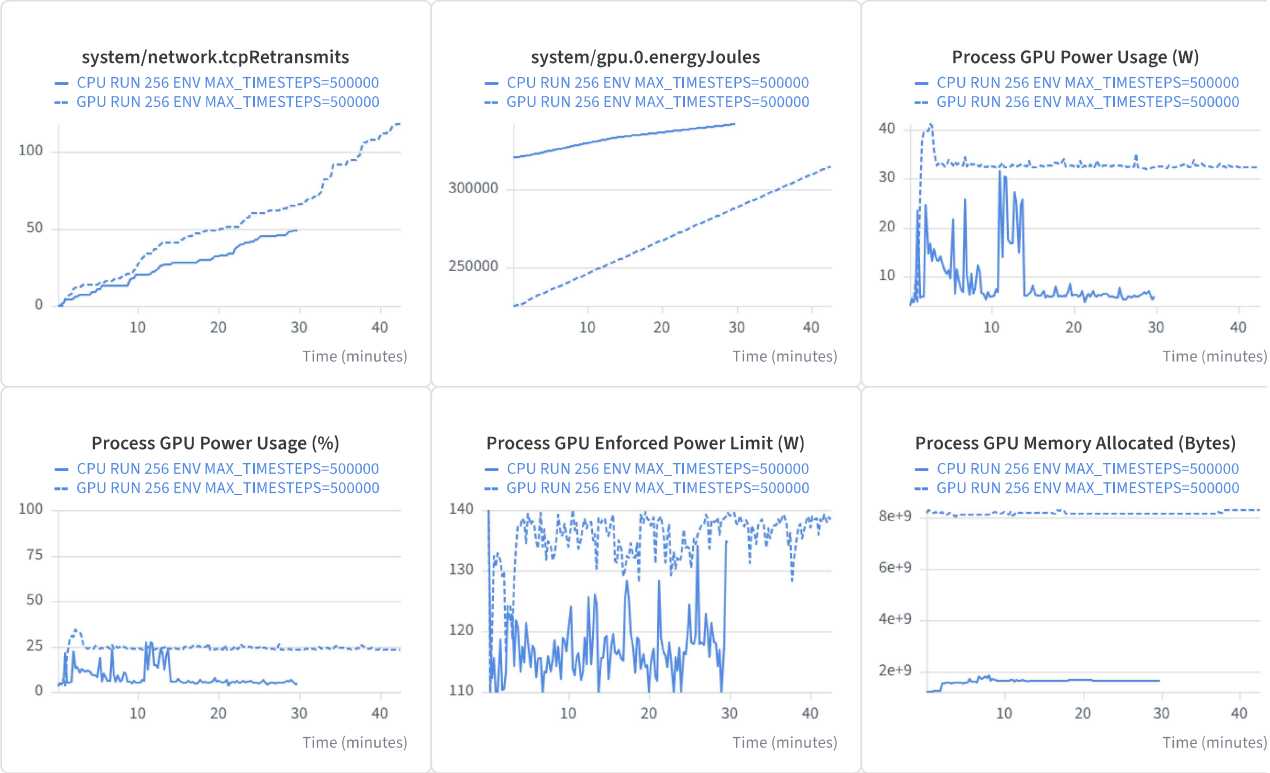
Add markdown, images, and $\LaTeX$

Add markdown, images, and $\LaTeX$

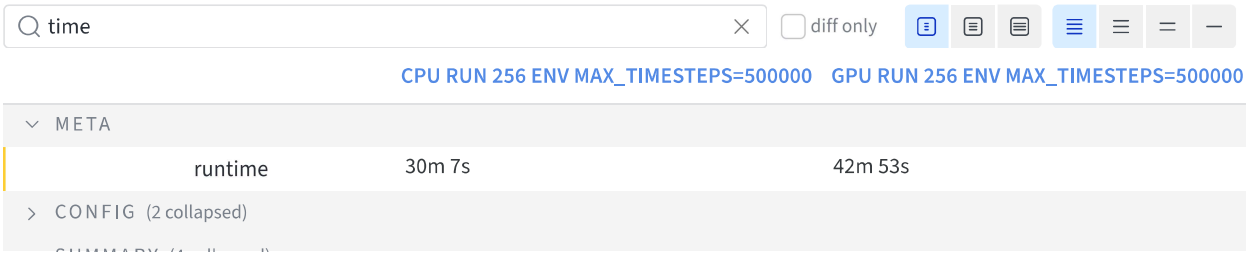
Add markdown, images, and $\LaTeX$



World-level metrics confirm environment determinism across hardware — step counter, order counter and episode time are identical in both runs. Market metrics (spread, time, window index) show higher variance at NUM_ENVS=256 compared to NUM_ENVS=1024, consistent with the statistical averaging benefit of larger parallel environment counts



Metric	GPU Run	CPU Run
Wall time	~45 min	~30 min
Energy (J)	~314,000 (still rising)	~341,000 (still rising)
GPU power (W)	~32 W (JIT spike)	~10 W (idle)
GPU power %	up to 33%	up to 27%
GPU memory	~8 GB	~1.8 GB



It is Clearly seen that run time is faster for CPU RUN ENV =256 Than that of GPU's